

YANCHENG ZHOU

Homepage $\diamond$ GitHub $\diamond$ LinkedIn

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EDUCATION

University of California, Davis (UCD)

May 2027 (expected)

M.S. in Statistics

GPA: 3.9/4.0

Related courses: High dimensional Statistics, Appl & Comp Statistics, Time Series Analysis.

Shanghai Jiao Tong University

June 2025

B.E. in Mathematics and Applied Mathematics

Junior GPA: 3.6/4.3

Related courses: Functional Analysis, Differential Geometry, Stochastic Process, ODE, PDE, Numeric Methods of ODE and PDE, Topology.

RESEARCH INTERESTS

My research interests lie in machine learning theory, optimal transport, and computational neuroscience. More broadly, I am interested in using mathematical tools such as high-dimensional statistics, functional analysis, and differential geometry to understand complex learning systems.

For machine learning theory, I am interested in the dynamical mechanisms of optimization methods, especially the oscillatory behaviors of gradient descent near the edge of stability. My current interest is in understanding how edge-of-stability dynamics, implicit regularization, and optimization geometry can be used to design improved gradient-based algorithms.

For optimal transport and Wasserstein space, I am interested in the limit behavior and dynamical properties of optimal transport maps, and the geometric structures induced by Wasserstein metrics.

For computational neuroscience, I am interested in mathematical descriptions of non-equilibrium neural networks and principled quantitative estimates of coding quality in neural systems.

RESEARCH EXPERIENCE

Limit Behavior of Optimal Transport

Feb 2026 – Present

Supervisor: Prof. Hans-Georg Müller

UC Davis

- Studied the asymptotic behavior of iterated optimal transport maps in Wasserstein space from the perspective of transport dynamics and Otto calculus.
- In the one-dimensional setting, characterized limiting points through intersections of cumulative distribution functions and analyzed the stability of fixed points.
- In higher dimensions, analyzed the Monge–Ampère equation and proved local stability results for transport fixed points under perturbations of target densities.
- Developed Sinkhorn-based numerical simulations to verify theoretical stability results on bounded domains.

Encoding of Non-balanced Networks in the Visual Cortical System March 2024 – June 2025

Supervisor: Prof. Dongzhao Zhou

Shanghai, China

- Studied theoretical foundations including mean-field approximation and the Fokker–Planck equation.
- Modeled visual cortical neuron encoding processes using Brian2 and reconstructed neural information with U-Net architectures.
- Investigated the effects of non-balanced neural networks on firing patterns and information encoding dynamics.

Edge-of-Stability Dynamics in Gradient Descent

Independent Research Project

Spring 2026 – Present

UC Davis

- Studied the self-stabilization mechanism of gradient descent near the edge of stability (EOS) using projected gradient dynamics and higher-order Taylor expansions.
- Reconstructed reduced oscillatory dynamics through Hessian eigendirection and sharpness-gradient decomposition, explaining the negative-feedback mechanism near the EOS threshold.
- Implemented CIFAR10 experiments reproducing sharpness oscillations and phase portrait dynamics predicted by theory.
- Investigated whether oscillation decay can be interpreted as a dissipative effect of gradient descent and explored improved optimization algorithms from symplectic perspectives.

SELECTED PROJECTS

Exploration of Quantum DeepONet

Sept. 2024 – Jan. 2025

- Studied theoretical foundations of DeepONet, including universal approximation theorems for operators and error analysis techniques based on covariance operator decomposition.
- Developed a Quantum DeepONet framework using quantum neural networks as branch networks and analyzed implicit frequency bias associated with network depth.
- Designed additional linear frequency-capturing layers achieving comparable performance to classical DeepONet with significantly fewer parameters.

Symplectic Geometric Algorithms for Hamiltonian Systems

June 2024

- Investigated symplectic geometric algorithms for Hamiltonian systems to preserve energy and reduce numerical dissipation.
- Evaluated the accuracy of symplectic integrators for approximating Hamiltonian dynamics.
- Explored applications of symplectic schemes in Hamiltonian Neural Networks (HNNs).

Exploring Dendritic Computation: Bilinear Rule and Beyond

Jan. 2024

Supervisor: Prof. Dongzhao Zhou

- Studied dendritic computation mechanisms in biological neurons using asymptotic analysis of Hodgkin–Huxley equations.
- Derived and verified bilinear interaction rules through NEURON simulations.
- Constructed hLN and dendritic bilinear neural network models for neuronal dynamics.

SKILLS/HOBBIES

Programming Languages

Python, C/C++, MATLAB, Java, Javascript

Machine Learning Tools

Pytorch, Tensorflow, Sklearn, Pandas, Numpy

Hobbies

Basketball, Swimming

HONORS AND AWARDS

Outstanding Master’s Student Award, UC Davis Statistics (2026)

Zhiyuan Honors Scholarship for Academic Excellence (2021–2022)

1st Prize, SJTUMCM Competition – Airplane Path Planning and Control (Dec. 2022)

3rd Prize in Shanghai District, National Mathematical Contest in Modeling (Jun. 2022)